

Forecast-available signals

311 history + calendar
Weather at t + IDs



Leakage-controlled panel

NTA x week x category
history $t \rightarrow$ event $t+1$

$t \rightarrow t+1$ design



Prospective evaluation

LightGBM + XGBoost
2015-22 train | 2023 val.
2024-25 test



Category-calibrated alerts

ensemble score
service threshold
risk alert + SHAP review

validation only

OSM + PLUTO 2026: retrospective context only

excluded from prospective alerts